

Determine a and b such that the integral

$$\int_0^\pi (x - a \sin(x) - b \sin(2x))^2 dx$$

is minimized.

Solution:

$$v \in \mathcal{M} = [\sin(x), \sin(2x)]$$

$\sin(x)$ and $\sin(2x)$ are orthogonal and we can use the projection formula which says that the minimum is obtained when

$$v = P_{[\mathcal{M}]} u$$

$$v_1 = a \sin(x) = P_{[\sin(x)]} u = \frac{(\sin(x) | x)}{(\sin(x) | \sin(x))} \sin(x)$$

$$v_2 = b \sin(2x)$$

From this, a is calculated as

$$a = \frac{(\sin(x) | x)}{(\sin(x) | \sin(x))}$$

$$(\sin(x) | x) = \int_0^\pi x \sin(x) dx = -[x \cos(x)]_0^\pi + \int_0^\pi \pi \cos(x) dx = \pi$$

$$(\sin(x) | \sin(x)) = \int_0^\pi \sin^2 x dx = \int_0^\pi \frac{1 - \cos(2x)}{2} dx = \frac{\pi}{2}$$

↓

$$a = \frac{\pi}{\frac{\pi}{2}} = 2$$

and b is calculated in the same way as

$$b = \frac{(\sin(2x) | x)}{(\sin(2x) | \sin(2x))}$$

$$(\sin(2x) | x) = \int_0^\pi x \sin(2x) dx = -\left[\frac{x \cos(2x)}{2}\right]_0^\pi + \int_0^\pi \frac{\cos(2x)}{2} dx = -\frac{\pi}{2}$$

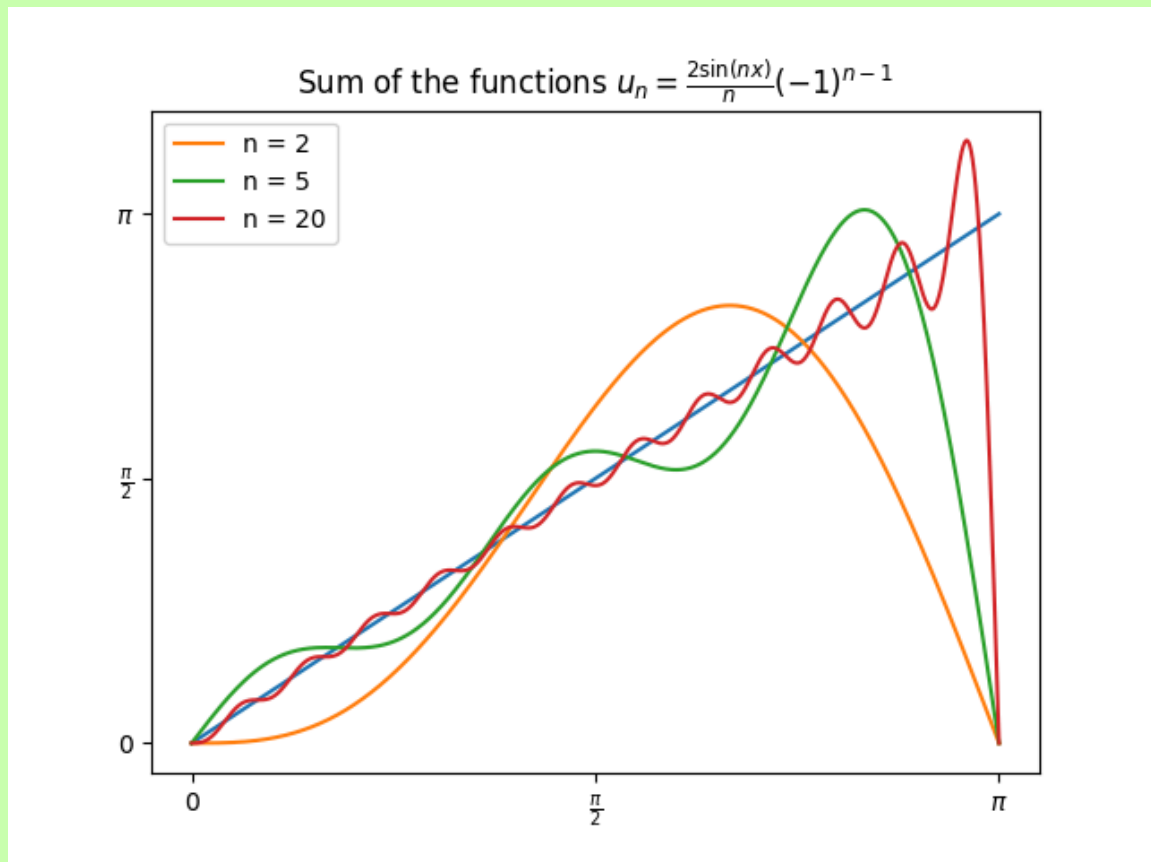
$$(\sin(2x) | \sin(2x)) = \int_0^\pi \sin^2(2x) dx = \int_0^\pi \frac{1 - \cos(2x)}{2} dx = \frac{\pi}{2}$$

↓

$$b = -\frac{\frac{\pi}{2}}{\frac{\pi}{2}} = -1$$

In general, the functions $v \in \mathcal{M} = \{\sin(nx)\}_{n=1}^\infty$ that best approximate $u(x) = x$ at the interval $[0, \pi]$ is given by

$$u_n = \frac{2(-1)^{n-1} \sin nx}{n}$$



□